EVERY FPPF COVERING IS REFINED BY A TWO-STAGE ZARISKI-THEN-FINITE-LOCALLY-FREE TOWER

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 13648-13660 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 13662-13683, the complete author proof (20 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Third member of the fppf-domination trio completed with this wave (L16107 etale-local, L16118 finite-global, this Zariski-tower form); all three distinct aggregate statements.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemma dominate-fppf-etale-locally (card L16107, on the shelf); the chapter's remark-topologies approximation of etale coverings by Zariski towers with finite locally free pieces; stability of finite locally free under base change.

Source reference. Original lemma 0DET: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13608–13614. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02LI: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11874–11892. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $\{S_i \to S\}_{i \in I}$ be an fppf covering. Then there exist*

(1) *a Zariski open covering $S = \bigcup U_j$,*
(2) *surjective finite locally free morphisms $W_j \to U_j$,*
(3) *Zariski open coverings $W_j = \bigcup_k W_{j,k}$,*
(4) *surjective finite locally free morphisms $T_{j,k} \to W_{j,k}$*

such that the fppf covering $\{T_{j,k} \to S\}$ refines the given covering $\{S_i \to S\}$.

Proof. Let $\{V_a \to S\}_{a \in A}$ be the fppf covering found in Lemma 0DET. In other words, this covering refines $\{S_i \to S\}$ and each $V_a \to S$ factors as $V_a \to S'_a \to S$ with $S'_a \to S$ étale and $V_a \to S'_a$ surjective finite locally free.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

By Remark 02LI there exists a Zariski open covering $S = \bigcup U_j$, for each j a finite locally free, surjective morphism $W_j \to U_j$, and for each j a Zariski open covering $\{W_{j,k} \to W_j\}$ such that the family $\{W_{j,k} \to S\}$ refines the étale covering $\{S'_a \to S\}$, i.e., for each pair j, k there exists an $a(j,k)$ and a factorization $W_{j,k} \to S'_a \to S$ of the morphism $W_{j,k} \to S$. Set $T_{j,k} = W_{j,k} \times_{S'_a} V_a$ and everything is clear. $\square$